

Statistics

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What is Sampling?

- **Population:** The entire collection of individuals or items under study.
- **Sample:** A subset of the population selected for analysis.
- **Sampling:** The process of selecting a portion (sample) from the population to draw conclusions about the entire population.

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Example

A college has 5000 students. A researcher wants to estimate the average GPA. Instead of all 5000, she selects 200 students randomly — this 200 is the **sample**.

Why Sampling is Needed?

- **Cost-effective:** Studying the entire population is expensive.
- **Time-saving:** Easier and faster than a census.
- **Feasibility:** Some populations are infinite or inaccessible.
- **Accuracy:** Proper sampling gives reliable results with manageable error.

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- **Accuracy:** Proper sampling gives reliable results with manageable error.

Example

Testing the strength of steel rods — we cannot test every rod; instead, we test a few to infer the quality.

- **Sampling Unit:** The individual element (e.g., a student, machine, house).
- **Sampling Frame:** A list of all sampling units in the population.
- **Parameter:** A numerical summary of a population (e.g., population mean μ).
- **Statistic:** A numerical summary of a sample (e.g., sample mean $\bar{x}$).
- **Sampling Error:** Difference between a sample statistic and population parameter.

Classification of Sampling Methods

Probability Sampling

- Simple Random Sampling
- Systematic Sampling
- Stratified Sampling
- Cluster Sampling

Non-Probability Sampling

- Convenience Sampling
- Judgment Sampling
- Quota Sampling
- Snowball Sampling

1. Simple Random Sampling (SRS)

- Each unit has an equal chance of being selected.
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Methods

- Lottery method
- Using random number tables or software

2. Systematic Sampling

- Every k th unit is selected from an ordered list.
- Sampling interval $k = \frac{N}{n}$ where N = population size, n = sample size.

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Example

Population of 100 students, want a sample of 10. Then $k = 100/10 = 10$. Pick one random start between 1 and 10, say 6. Then sample units: 6, 16, 26, 36, ..., 96.

3. Stratified Sampling

- Population divided into homogeneous groups called **strata**.
- A random sample is taken from each stratum.

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Example

Students divided into departments: CSE, ECE, ME, CE. Select 20 students randomly from each department.

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Advantages

Increases precision of estimates, ensures representation of all groups.

4. Cluster Sampling

- Population divided into clusters (heterogeneous groups).
- A few clusters are chosen randomly; all members in selected clusters are included.

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Example

A city divided into 100 blocks. Select 10 blocks randomly and survey all households within them.

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Note

Cheaper but less precise compared to stratified sampling.

1. Convenience Sampling

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Surveying friends or students available in the lab.

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Limitation: Highly biased; not representative.

2. Judgment and Quota Sampling

- **Judgment Sampling:** Researcher selects samples based on their expertise.
- **Quota Sampling:** Researcher ensures fixed numbers from each group (e.g., 50% male, 50% female respondents).

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Example

An interviewer selects 10 experienced engineers and 10 fresh graduates for a study.

3. Snowball Sampling

- Used when population is hard to locate.
- Existing subjects recruit future subjects among their acquaintances.

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Research on rare disease patients.

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Drawback: Lacks randomness; prone to bias.

Sampling Error and Bias

- **Sampling Error:** Difference between the sample statistic and population parameter due to chance.
- **Non-sampling Error:** Errors due to faulty design, recording, or non-response.

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Example

If population mean height = 165 cm, and sample mean = 163 cm, then sampling error = 2 cm.

What is an Estimator?

- **Population parameter:** A fixed but unknown numerical characteristic of the population, e.g. mean μ , variance σ^2 , proportion p .
- **Estimator:** A rule or formula used to estimate the parameter from sample data.
- **Estimate:** The numerical value obtained after substituting the sample observations into the estimator.

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Notation

If $\hat{\theta}$ is an estimator of parameter θ , then

$$\hat{\theta} = T(X_1, X_2, \dots, X_n)$$

where T is a statistic based on the random sample.

Examples of Estimators

- Estimator of population mean: $\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$
- Estimator of population variance: $\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$
- Estimator of population proportion: $\hat{p} = \frac{x}{n}$ where x = number of successes

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Example

Suppose we have a sample of exam marks: 40, 60, 50, 70, 80. Then $\bar{X} = 60$ is the **estimate** of the population mean.

Properties of Good Estimators

A good estimator should possess the following properties:

- ① Unbiasedness
- ② Consistency
- ③ Efficiency
- ④ Sufficiency
- ⑤ Minimum Variance (MVUE)

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These properties ensure that estimators are reliable and close to the true parameter values.

1. Unbiasedness

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$$E(\hat{\theta}) = \theta$$

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Example

Let $X_1, X_2, \dots, X_n$ be a random sample from a population with mean μ .
Estimator: $\bar{X} = \frac{1}{n} \sum X_i$.

$$E(\bar{X}) = \frac{1}{n} \sum E(X_i) = \mu \Rightarrow \bar{X} \text{ is unbiased for } \mu.$$

Example of Biased Estimator

- Estimator of population variance using $\frac{1}{n} \sum (X_i - \bar{X})^2$ is **biased**.
- Its expectation is $\frac{n-1}{n} \sigma^2$.

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Correction

Using $\frac{1}{n-1} \sum (X_i - \bar{X})^2$ gives an **unbiased estimator** of σ^2 .

2. Consistency

- An estimator $\hat{\theta}_n$ is **consistent** if it converges in probability to the true parameter as $n \rightarrow \infty$.

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Example

Sample mean $\bar{X}$ is a consistent estimator of population mean μ , since

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$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Interpretation: Larger samples yield more accurate estimates.

3. Efficiency

- Among all unbiased estimators, the one with the smallest variance is called the **most efficient estimator**.
- If $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased for θ , and

$$\text{Var}(\hat{\theta}_1) < \text{Var}(\hat{\theta}_2),$$

then $\hat{\theta}_1$ is more efficient than $\hat{\theta}_2$.

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Example

For a normal population, both $\bar{X}$ (sample mean) and sample median estimate μ . But $\text{Var}(\bar{X}) < \text{Var}(\text{median})$, hence $\bar{X}$ is more efficient.

4. Sufficiency

- A statistic $T(X_1, \dots, X_n)$ is **sufficient** for θ if it contains all the information about θ available in the sample.
- Mathematically, T is sufficient if

$$f(x_1, x_2, \dots, x_n | T) = g(x_1, x_2, \dots, x_n)$$

does not depend on θ .

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Example

For a normal distribution $N(\mu, \sigma^2)$ with known σ , $\bar{X}$ is a sufficient statistic for μ .

5. Minimum Variance Unbiased Estimator (MVUE)

- An estimator that is both **unbiased** and has the **smallest variance** among all unbiased estimators.
- Often obtained using:
 - Rao–Blackwell Theorem
 - Lehmann–Scheffé Theorem

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Example

For normal population $N(\mu, \sigma^2)$,

$\bar{X}$ is MVUE of μ .

Example: Comparing Two Estimators

Let X_1, X_2, X_3 be independent and identically distributed (i.i.d) random variables with mean μ and variance σ^2 .

$$\hat{\mu}_1 = \frac{X_1 + X_2 + X_3}{3}, \quad \hat{\mu}_2 = \frac{X_1 + 2X_2}{3}$$

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Compare:

- Both are unbiased: $E(\hat{\mu}_1) = E(\hat{\mu}_2) = \mu$
- Variances:

$$\text{Var}(\hat{\mu}_1) = \frac{\sigma^2}{3}, \quad \text{Var}(\hat{\mu}_2) = \frac{5\sigma^2}{9}$$

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- Variances:

$$Var(\hat{\mu}_1) = \frac{\sigma^2}{3}, \quad Var(\hat{\mu}_2) = \frac{5\sigma^2}{9}$$

Conclusion: $\hat{\mu}_1$ is more efficient $\Rightarrow$ preferred estimator.

What is Hypothesis Testing?

- A **statistical hypothesis** is an assumption about a population parameter.
- **Hypothesis testing** helps decide whether sample data supports or contradicts that assumption.
- Used widely in data analytics, AI model validation, and performance benchmarking.

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Example

A researcher claims a new search algorithm reduces average query response time below 0.4 s. We test this claim using hypothesis testing.

General Procedure

- 1 State the **Null hypothesis** (H_0) and **Alternative hypothesis** (H_1).
- 2 Select significance level α (commonly 0.05 or 0.01).
- 3 Choose suitable test statistic (Z, t, Chi-square).
- 4 Compute test statistic from sample data.
- 5 Find critical value or p-value.
- 6 Draw conclusion:
 - If test statistic lies in rejection region $\rightarrow$ Reject H_0 .
 - Else $\rightarrow$ Fail to reject H_0 .

Types of Hypotheses

- **Null hypothesis (H_0):** No difference or no effect.
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Examples

- H_0 : Average CPU response time = 200 ms.
 H_1 : Average CPU response time < 200 ms.
- H_0 : Mean accuracy of two ML models is equal.
 H_1 : Mean accuracy of models differs.

Types of Errors

- **Type I error:** Rejecting H_0 when it is true (false alarm) (false positive). Probability = α .
- **Type II error:** Failing to reject H_0 when it is false (missed detection) (false negative). Probability = β .
- α is the **significance level**.
- $1 - \beta$ is the **Power of the test**.

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- α is the **significance level**.
- $1 - \beta$ is the **Power of the test**.

Example

Flagging a network as “anomalous” when it’s actually normal → Type I error. Failing to detect an attack → Type II error.

Significance Level (α)

Definition

The **significance level** (α) is the **probability of rejecting the null hypothesis** (H_0) **when it is actually true**.

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- It represents the **risk of making a Type I error.**
- Common choices of α are:

0.10, 0.05, 0.02, 0.01

- A smaller α means we require **stronger evidence** to reject H_0 .

Example

If $\alpha = 0.05$, there is a **5% chance** of rejecting H_0 even though it is true.

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Interpretation

Significance level defines the **boundary between acceptance and rejection** of H_0 .

If $p \leq \alpha \Rightarrow$ Reject H_0 , else Do not reject H_0

Definition

The **power of a test** is the probability that the test **correctly rejects the null hypothesis** (H_0) when the alternative hypothesis (H_1) is true.

$$\text{Power} = P(\text{Reject } H_0 \mid H_1 \text{ is true})$$

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- It measures the **sensitivity** of a statistical test.
- A test with high power is more likely to detect an actual effect or difference.
- Power is related to the **Type II error** (β) as:

$$\text{Power} = 1 - \beta$$

Example

If $\beta = 0.20$, then the power of the test is:

$$1 - 0.20 = 0.80$$

This means the test has an 80% chance of correctly rejecting H_0 when H_1 is true.

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Factors Affecting Power

- Sample size (n)
- Significance level (α)
- True difference between parameters
- Variability in data

Two-Tailed Test

Purpose: Test if a parameter is **different** from a hypothesized value, without specifying direction.

Hypotheses:

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_1 : \mu \neq \mu_0$$

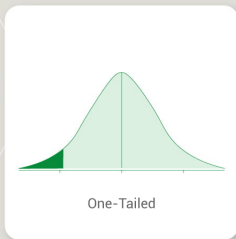
Critical region: Split on both tails of the distribution. Each tail has significance level $\alpha/2$.

Decision rule: Reject H_0 if

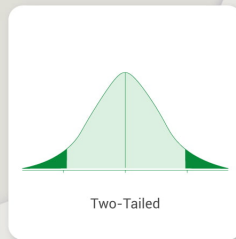
$$|t| > t_{n-1, \alpha/2}$$

Example: Machine fills bottles with 500 ml. Test if the mean fill **differs** from 500 ml.

Difference between **One-Tailed** and **Two-Tailed** Hypothesis



VS



- Shaded regions on both ends represent $\alpha/2$ each.
- Reject H_0 if t falls in either tail.

One-Tailed Test

Purpose: Test if a parameter is **greater than or less than** a hypothesized value (specific direction).

Hypotheses:

- Right-tailed: $H_0 : \mu = \mu_0, H_1 : \mu > \mu_0$
- Left-tailed: $H_0 : \mu = \mu_0, H_1 : \mu < \mu_0$

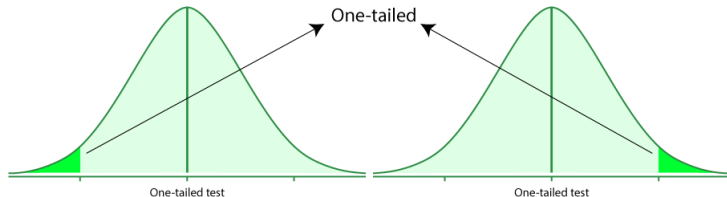
Critical region: Entire α significance level is in one tail.

Decision rule:

$$t > t_{n-1,\alpha} \quad (\text{right-tailed}), \quad t < -t_{n-1,\alpha} \quad (\text{left-tailed})$$

Example: Test if a new fertilizer **increases** yield above 50 kg per plot.

One-Tailed Test: Diagram



- Right-tail test: shaded region on the right represents α .
- Left-tail test: shaded region on the left.
- Reject H_0 if t falls in shaded region.

Comparison of One-Tailed and Two-Tailed Tests

Feature	Two-Tailed	One-Tailed
Purpose	Test for any difference	Test for specific direction
H1	$\mu \neq \mu_0$	$\mu > \mu_0$ or $\mu < \mu_0$
Critical region	Split in two tails, $\alpha/2$ each	Entire α in one tail
Example	Mean differs from 500 ml	Mean $>$ 500 ml

Confidence Interval (CI)

A **confidence interval** gives a range of possible values for a population parameter (mean, proportion, etc.) based on sample data.

Confidence Intervals: One-Sample and Two-Sample Tests

Test	Sample Type	Parameter	Confidence Interval (CI)
Z-test	One-sample	Mean (μ)	$\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
Z-test	Two-sample	Difference of means ($\mu_1 - \mu_2$)	$(\bar{X}_1 - \bar{X}_2) \pm Z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$
t-test	One-sample	Mean (μ)	$\bar{X} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$
t-test	Two-sample	Difference of means ($\mu_1 - \mu_2$)	$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2, df} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
Chi-square	One-sample	Variance (σ^2)	$\left(\frac{(n-1)s^2}{\chi_{\alpha/2, n-1}^2}, \frac{(n-1)s^2}{\chi_{1-\alpha/2, n-1}^2} \right)$

Z-Test for Mean(s)

Purpose: To test hypotheses about population mean(s) when the population standard deviation(s) are known.

Case 1: One-Sample Z-Test

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

Used when:

- Population standard deviation σ is known.
- Sample size n is large ($n \geq 30$) or population is normal.
- Z : Z-statistic (standardized test statistic)
- $\bar{X}$: Sample mean
- μ_0 : Population mean under the null hypothesis (H_0)
- σ : Population standard deviation
- n : Sample size

Case 2: Two-Sample Z-Test

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Used when:

- Both population standard deviations σ_1, σ_2 are known.
 - Two samples are independent.
-
- Z : Z-statistic for two-sample test
 - $\bar{X}_1$: Sample mean of the first sample
 - $\bar{X}_2$: Sample mean of the second sample
 - σ_1 : Population standard deviation of the first sample
 - σ_2 : Population standard deviation of the second sample
 - n_1 : Sample size of the first sample
 - n_2 : Sample size of the second sample

Critical Z-values for Different Tests

Significance Level (α)	Two-tailed Test ($\alpha/2$ each)	Right-tailed Test	Left-tailed Test
10% (0.10)	± 1.64	+1.28	-1.28
5% (0.05)	± 1.96	+1.645	-1.645
2% (0.02)	± 2.33	+2.05	-2.05
1% (0.01)	± 2.58	+2.33	-2.33

Note:

- For a **two-tailed test**, the rejection regions lie in both tails, each with area $\alpha/2$.
- For a **right-tailed test**, rejection occurs when $Z > Z_{\alpha}$.
- For a **left-tailed test**, rejection occurs when $Z < -Z_{\alpha}$.

One-Sample Z-Test: Problem 1

A company manufactures resistors with a **claimed mean resistance of 100 Ω** . The population standard deviation is $\sigma = 2.5 \Omega$.

A random sample of $n = 60$ resistors is taken, with sample mean $\bar{X} = 99.4 \Omega$.

Tasks:

- 1 At $\alpha = 0.01$, test if the true mean differs from 100 Ω .
- 2 Construct the 99% confidence interval for μ .
- 3 Interpret the result in terms of production accuracy.

One-Sample Z-Test: Problem 2

A coffee shop claims that the average amount of coffee in its cups is **250 ml**. A random sample of **36** cups is measured, and the sample mean is found to be **247 ml**. The population standard deviation is known: $\sigma = 5 \text{ ml}$.

One-Sample Z-Test: Problem 2

A coffee shop claims that the average amount of coffee in its cups is **250 ml**. A random sample of **36** cups is measured, and the sample mean is found to be **247 ml**. The population standard deviation is known: $\sigma = 5 \text{ ml}$.

Tasks:

- 1 At $\alpha = 0.05$, test whether the average amount differs from 250 ml.
- 2 Construct a 95% confidence interval for the mean coffee amount.
- 3 Interpret the result in the context of the coffee shop's claim.

Two-Sample Z-Test: Problem 3

Two battery brands, **Brand A** and **Brand B**, are claimed to have equal mean lifetimes. Population standard deviations are known: $\sigma_A = 3.5$ hrs, $\sigma_B = 4.0$ hrs.

Samples:

- Brand A: $n_1 = 40$, $\bar{X}_1 = 41.8$ hrs
- Brand B: $n_2 = 50$, $\bar{X}_2 = 39.6$ hrs

Tasks:

- 1 At $\alpha = 0.01$, test if the mean lifetimes differ significantly.
- 2 Construct a 95% confidence interval for $(\mu_A - \mu_B)$.
- 3 Discuss implications if population variances were unknown or unequal.

Two-Sample Z-Test: Problem 4

A simple random sampling survey in respect of monthly earnings of semi-skilled workers in two cities gives the following statistical information.

City	Mean monthly earnings (Rs)	SD of sample data of monthly earnings	Size sample of
A	695	40	200
B	710	60	175

Test the hypothesis at 5% significance level that there is no difference between monthly earnings of workers in two cities.

Problem 5

A factory claims that its light bulbs last **at least 1200 hours** on average. A random sample of **40** bulbs is tested, and the sample mean lifetime is **1185 hours**. The population standard deviation is known: $\sigma = 30$ hours.

Tasks:

- 1 At $\alpha = 0.05$, test whether the mean lifetime is less than 1200 hours.
- 2 Determine if the factory's claim is supported by the sample.

Summary: One-Sample vs Two-Sample Z-Tests

Feature	One-Sample Z-Test	Two-Sample Z-Test
Purpose	Compare sample mean with population	Compare two sample means or proportions
Population σ	Known	Both σ_1, σ_2 known
Null Hypothesis	$H_0 : \mu = \mu_0$	$H_0 : \mu_1 = \mu_2$
Critical Value	Z_α	$Z_{\alpha/2}$
Decision	Reject if $ Z > Z_\alpha$	Reject if $ Z > Z_{\alpha/2}$

t-Test: Introduction

- Used to compare sample means.
- Assumes data is approximately normally distributed.
- Types of t-tests:
 - One-sample t-test
 - Two-sample t-test (independent)
 - Paired t-test

One-Sample t-Test

Purpose: Compare the mean of a single sample with a known population mean μ_0 .

Hypotheses:

$$H_0 : \mu = \mu_0, \quad H_1 : \mu \neq \mu_0$$

Test statistic:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

- $\bar{X}$: sample mean
- s : sample standard deviation
- n : sample size

Decision rule: Compare $|t|$ with $t_{n-1, \alpha/2}$ from t-table.

One-Sample t-Test: Example

- Sample: $n = 10$ students, average score $\bar{X} = 78$, $s = 5$
- Population mean $\mu_0 = 75$
- $t = \frac{78-75}{5/\sqrt{10}} = \frac{3}{1.581} \approx 1.897$
- Degrees of freedom: $n - 1 = 9$
- Compare with $t_{0.025,9} = 2.262$
- Since $1.897 < 2.262$, fail to reject H_0

Two-Sample t -Test Overview

Two-sample t -tests compare means of **two independent groups**. Key consideration: Are **population variances equal or unequal?**

1. Equal Variances (Pooled t -Test)

Used when: Population variances are unknown but assumed equal.

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Pooled standard deviation:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Degrees of freedom:

$$df = n_1 + n_2 - 2$$

2. Unequal Variances (Welch's t -Test)

Used when: Population variances are unknown and not assumed equal.

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Degrees of freedom (Welch–Satterthwaite approximation):

$$df \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}$$

Two-Sample t-Test: Example

- Sample 1: $n_1 = 12, \bar{X}_1 = 80, s_1 = 4$
- Sample 2: $n_2 = 10, \bar{X}_2 = 75, s_2 = 5$
- Pooled variance:

$$s_p^2 = \frac{(11)(16) + (9)(25)}{20} = \frac{176 + 225}{20} = 20.05$$

- $t = \frac{80-75}{\sqrt{20.05}\sqrt{\frac{1}{12} + \frac{1}{10}}} \approx 2.145$
- Degrees of freedom: $n_1 + n_2 - 2 = 20$
- Compare with $t_{0.025,20} = 2.086 \Rightarrow \text{reject } H_0$

TABLE VIII : CRITICAL VALUES OF STUDENT'S t-DISTRIBUTION

d.f.	LEVEL OF SIGNIFICANCE FOR ONE-TAILED TEST						
	.25	.10	.05	0.025	.01	.005	.0005
v	LEVEL OF SIGNIFICANCE FOR TWO-TAILED TEST						
	.50	.20	.10	.05	.02	.01	.001
1	1.000	3.078	6.314	12.706	31.821	63.657	636.619
2	.816	1.886	2.920	4.303	6.965	9.925	31.599
3	.765	1.638	2.353	3.182	4.541	5.841	12.924
4	.741	1.533	2.132	2.776	3.747	4.604	8.610
5	.727	1.476	2.015	2.571	3.365	4.032	6.869
6	.718	1.440	1.943	2.447	3.143	3.707	5.959
7	.711	1.415	1.895	2.365	2.998	3.499	5.408
8	.706	1.397	1.860	2.306	2.896	3.355	5.041
9	.703	1.383	1.833	2.262	2.821	3.250	4.781
10	.700	1.372	1.812	2.228	2.764	3.169	4.587
11	.697	1.363	1.796	2.201	2.718	3.106	4.437
12	.695	1.356	1.782	2.179	2.681	3.055	4.318
13	.694	1.350	1.771	2.160	2.650	3.012	4.221
14	.692	1.345	1.761	2.145	2.624	2.977	4.140
15	.691	1.341	1.753	2.131	2.602	2.947	4.073
16	.690	1.337	1.746	2.120	2.583	2.921	4.015
17	.689	1.333	1.740	2.110	2.567	2.898	3.965
18	.688	1.330	1.734	2.101	2.552	2.878	3.922
19	.688	1.328	1.729	2.093	2.539	2.861	3.883
20	.687	1.325	1.725	2.086	2.528	2.845	3.850
21	.686	1.323	1.721	2.080	2.518	2.831	3.819
22	.686	1.321	1.717	2.074	2.508	2.819	3.792
23	.685	1.319	1.714	2.069	2.500	2.807	3.768
24	.685	1.318	1.711	2.064	2.492	2.797	3.745
25	.684	1.316	1.708	2.060	2.485	2.787	3.725
26	.684	1.315	1.706	2.056	2.479	2.779	3.707
27	.684	1.314	1.703	2.052	2.473	2.771	3.690
28	.683	1.313	1.701	2.048	2.467	2.763	3.674
29	.683	1.311	1.699	2.045	2.462	2.756	3.659
30	.683	1.310	1.697	2.042	2.457	2.750	3.646
40	.681	1.303	1.684	2.021	2.423	2.704	3.551
60	.679	1.296	1.671	2.000	2.390	2.660	3.460
120	.677	1.289	1.658	1.980	2.358	2.617	3.373
∞	.674	1.282	1.645	1.960	2.326	2.576	3.291

Example 1 – Single Sample t-test (CPU Utilization)

A process scheduler claims mean CPU utilization = 70%. Sample of 10 runs: 68, 72, 69, 70, 65, 71, 68, 74, 69, 67. At $\alpha = 0.05$, verify claim.

Example 2 – Independent Samples t-test (Model Accuracy)

A researcher evaluates two ML models using 10-fold cross-validation:

- Model A: 91, 89, 90, 92, 88, 93, 90, 91, 89, 90
- Model B: 88, 85, 87, 86, 88, 89, 86, 87, 85, 86

Test whether there is a significant difference in mean accuracies at $\alpha = 0.05$. Assume the two samples are independent and variances are equal.

Example 3

A group of seven chickens reared on a high protein diet of 12, 15, 11, 16, 14, 14, and 16 ounces; a second group of five chickens of same age, similarly treated except that they receive a low protein diet, of 8, 10, 14, 10 and 13 ounces. Test at 5 per cent level whether there is significant evidence that additional protein has increased the weight of the chickens.

Comparison of Tests

Feature	Z-Test	t-Test	Chi-Square Test
Data Type	Quantitative	Quantitative	Categorical
Variance Known	Yes	No	Not required
Sample Size	Large	Small	Any
Purpose	Mean/Proportion	Mean	Frequencies
Distribution	Normal	Normal	Chi-square